

Problem 33 (Silk Road Mathematics Competition 2011 Problem 1). There are 5 sets such that the intersection of any two sets has exactly 1 element, and the intersection of any four sets is empty. Find the smallest possible number of elements in the union of these 5 sets.

Solution. The answer is 6.

Firstly, it is easy to show that the 5 sets below satisfy all conditions.

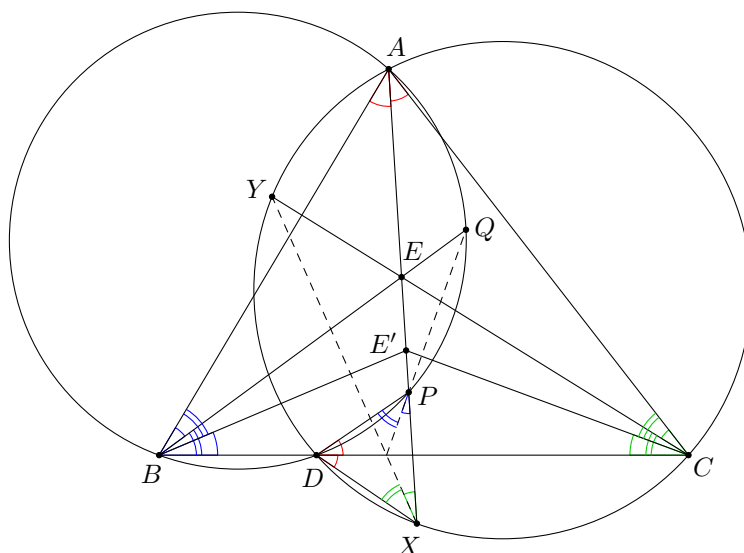
$$\{1, 2\}, \{1, 3, 4\}, \{1, 5, 6\}, \{2, 3, 5\}, \{2, 4, 6\}$$

There are 6 elements in the union.

It remains to show that it is impossible that the union consists of at most 5 elements. Suppose on the contrary that only 1, 2, 3, 4, 5 belong to some of the sets. Clearly, each element belongs to at most 3 sets. If there are at most 2 elements each belonging to exactly 3 sets, then each such element belongs to 3 pairs of sets, and each of the other elements belongs to at most 1 pair. Therefore, at most $3 \times 2 + 1 \times 3 = 9$ pairs of sets share a common element, which is a contradiction since there are $\binom{5}{2} = 10$ pairs.

Therefore, WLOG assume 1, 2, 3 each belongs to exactly 3 sets. Assume A_1, A_2, A_3 contain 1. Then 2 belongs to at most one of them, and so we may assume A_1, A_4, A_5 contains 2. But then 3 must belong to A_4, A_5 for the same reason, contradicting $|A_4 \cap A_5| = 1$. So it is impossible that the union contains at most 5 elements. The proof is complete. $\square$

Problem 34 (Kürschák Competition 2015 Problem 2). Let D be a point on the side BC of $\triangle ABC$. Let E be a point on the internal angle bisector of $\angle BAC$ which lies inside $\triangle ABC$. Suppose AE and BE meet the circumcircle of $\triangle ABD$ again at P and Q respectively (with $P \neq Q$). Suppose AE and CE meet the circumcircle of $\triangle ACD$ again at X and Y respectively (with $X \neq Y$). Prove that BC, PQ and XY are concurrent or parallel.



Solution. The result is obvious if D lies on AE (in which case $D = P = X$). WLOG assume $\angle BAD < \angle CAD$. Then P and X lie on different sides of the line BC .

Using directed angles, we have $\angle DPX = \angle DBA = \angle CBA$ and $\angle PDX = \angle ACD = \angle ACB$. This proves $\triangle DPX$ and $\triangle ABC$ are oppositely similar.

Let E' be the isogonal conjugate of E with respect to $\triangle ABC$. Since $\angle DPQ = \angle DBQ = \angle E'BA$, the lines PQ and BE' are corresponding lines under the similarity $\triangle DPX \sim \triangle ABC$. Similarly, since $\angle DXY = \angle DCY = \angle E'CA$, the lines XY and CE' are corresponding lines under the similarity $\triangle DPX \sim \triangle ABC$. Also, as $\angle XDC = \angle XAC = \angle BAX = \angle CDP$, the line

BC is an angle bisector of $\angle XDP$. Note that it must be the internal angle bisector since BC separates P and X . It follows that the lines DC' and AE' are corresponding lines under the similarity $\triangle DPX \sim \triangle ABC$.

Now, as BE', CE', AE' are concurrent at E' , the corresponding lines PQ, XY, DC are also concurrent. $\square$

Remarks. The use of the isogonal conjugate is not necessary as the result follows by Ceva's theorem. It seems like the lines cannot be parallel? I think they are parallel when E' is a point at infinity. This holds when E lies on (ABC) , but in that case E cannot lie inside $\triangle ABC$.

Problem 35 (Silk Road Mathematics Competition 2017 Problem 3). We choose 42 integers from 1 to 1000000 randomly. Prove that there exist 4 chosen numbers such that for any permutation (a, b, c, d) of them, we have $25(ab + cd)(ac + bd) \geq 16(ad + bc)^2$.

Solution. Let the integers be $x_1 < x_2 < \dots < x_{42}$. We claim that there exists k such that $x_{2k+1}x_{2k+2} < 4x_{2k-1}x_{2k}$. If not, then we have $x_{2k+1}x_{2k+2} \geq 4x_{2k-1}x_{2k}$ for $k = 1, 2, \dots, 20$, and so

$$x_{41}x_{42} \geq 4x_{39}x_{40} \geq 4^2x_{37}x_{38} \geq \dots \geq 4^{20}x_1x_2 > 4^{20} = 1048576^2 > (10^6)^2.$$

This is impossible.

Now, suppose $w < x < y < z$ are four of the chosen numbers such that $yz < 4wx$. We claim that they satisfy the desired result. Firstly, note that $wx + yz$ is larger than both $wy + xz$ and $wz + xy$ by the rearrangement inequality. Therefore, it suffices to prove

$$25(wy + xz)(wz + xy) \geq 16(wx + yz)^2.$$

Indeed, by the AM-GM inequality, we have

$$\frac{25(wy + xz)(wz + xy)}{16(wx + yz)^2} \geq \frac{25(2\sqrt{wxyz})^2}{16(wx + yz)^2} = \frac{25wxyz}{4(wx + yz)^2}.$$

To show that this is at least 1, it is the same as proving $4(wx)^2 - 17(wx)(yz) + 4(yz)^2 \leq 0$, which means $(wx - 4yz)(4wx - yz) \leq 0$. This is true since $wx < yz < 4wx$. So we are done. $\square$

Problem 36 (USA TST 2014 Test 2 Problem 3). Prove that for any positive integer N , there exists a prime p and a multiplicative subgroup G of $\mathbb{F}_p^\times = \{1, 2, \dots, p-1\}$ such that G has at least N elements, and there are no $x, y, z \in G$ (which can be equal) satisfying $x + y \equiv z \pmod{p}$. (A multiplicative subgroup of $\mathbb{F}_p^\times$ is of the form $\{a^k \pmod{p} : k \geq 0\}$ for some $a \in \mathbb{F}_p^\times$.)

Solution. Let q be any odd prime larger than N . Note that $f(x) = (x+1)^q - 1$ and $g(x) = x^q - 1$ are relatively prime. Indeed, the roots of $g(x)$ are 1 and the primitive q th roots of unity. If ω is such a root of unity, then its minimal polynomial is $x^{q-1} + x^{q-2} + \dots + 1$. But then $(x+1)^q - 1 = x(x^{q-1} + qx^{q-2} + \dots + q)$ is not a multiple of that. Therefore, we can find $P_1, Q_1 \in \mathbb{Q}[x]$ such that $f(x)P_1(x) + g(x)Q_1(x) = 1$. Thus, there exist $P_2, Q_2 \in \mathbb{Z}[x]$ and an integer c depending only on q such that $f(x)P_2(x) + g(x)Q_2(x) = c$.

Choose any odd prime p larger than c such that $p \equiv 1 \pmod{q}$. Such a prime exists by Dirichlet's theorem. Let g be a primitive root modulo p , and consider $a = g^{\frac{p-1}{q}} \in \mathbb{F}_p^\times$. Then the subgroup generated by a is $G = \{1, a, a^2, \dots, a^{q-1}\}$. If $a^x + a^y \equiv a^z \pmod{p}$, then $1 + s \equiv t \pmod{p}$ where $s = a^{y-x}$ and $t = a^{z-x}$.

Note that $s^q \equiv t^q \equiv 1 \pmod{p}$ since $s, t \in G$. This implies $s^q \equiv 1 \pmod{p}$ and $(s+1)^q \equiv 1 \pmod{p}$. In other words, we have $p \mid f(s)$ and $p \mid g(s)$. So $p \mid f(x)P_2(x) + g(x)Q_2(x) = c$, contradicting $p > c$.

Therefore, it is not possible that $a^x + a^y \equiv a^z \pmod{p}$. This proves G is a subgroup of $q > N$ elements satisfying the given conditions. $\square$